Yeeun Shin

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RESEARCH INTERESTS

Human-Computer Interaction (HCI) • Extended Reality (XR) • Human-AI Interaction • Embodied and Multimodal Interaction • Adaptive and Agentic Interfaces

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) *Mar. 2021 – Aug. 2023*
M.S. in Industrial Design (Human-Computer Interaction) *Daejeon, Korea*
Minor in Entrepreneurship and Innovation
- Thesis: *Immersive Authoring Interface with 3D Virtual Controls on Physical Desk*
- Advisor: Woohun Lee | Committee: Seok-Hyung Bae, Andrea Bianchi

Pohang University of Science and Technology (POSTECH) *Mar. 2016 – Feb. 2021*
B.S. in Materials Science and Engineering *Pohang, Korea*
- Graduated *Cum Laude*

Institut National des Sciences Appliquées de Lyon (INSA Lyon) *Aug. 2019 - Jan. 2020*
Exchange Student, Materials Science and Engineering *Villeurbanne, France*

PUBLICATIONS

[6] Spatial Acts as Prompts: Embodied Prompting for AI-Assisted Sensemaking in XR
Yeeun Shin, Chenyang Zhang, Tianjian Wei, Nels Numan, Eric J Gonzalez, Mar Gonzalez-Franco, Yalong Yang, *(Under Review · Full Paper)*

[5] Desk Console: Augmenting 3D Virtual Controls on Physical Desks for Immersive Authoring
Yeeun Shin, Seung Hyeon Han, Woohun Lee
ACM CHI Conference on Human Factors in Computing Systems (CHI '25 Interactivity)

[4] Spatial Chef: A Spatial Transforming VR Game with Full Body Interaction
Yeeun Shin*, Yewon Lee*, Sungbaek Kim*, Soomin Park*
ACM CHI Conference on Human Factors in Computing Systems (CHI '23 SGC)
🏆 Student Game Competition Finalist

[3] WonderScope: Practical Near-surface AR Device for Museum Exhibits
HyeonBeom Yi, **Yeeun Shin**, Sehee Lee, Eunhye Youn, Auejin Ham, Geehyuk Lee, Woohun Lee
ACM SIGGRAPH 2022 Emerging Technologies
🏆 Best in Show Honorable Mention

[2] ProjecString: Turning an Everyday String Curtain Into an Interactive Projection Display
Wooje Chang*, **Yeeun Shin***, Yeon Soo Kim*, Woohun Lee
ACM SIGGRAPH 2022 Posters

[1] ChromoFilament: Designing a Thermochromic Filament for Displaying Malleable States
Donghyeon Ko, **Yeeun Shin**, Junbeom Shin, Jiwoo Hong, Woohun Lee
ACM Designing Interactive Systems Conference (DIS '22)

** denotes equal contributions.*

AWARDS & HONORS

- **CMU XRTC Creative Research Grant** | Carnegie Mellon University *2026*
- **ACM CHI Student Game Competition Finalist** *2023*
- **ACM SIGGRAPH Best in Show Honorable Mention** *2022*
- **Edison Award Gold Winner**, Samsung Vision AI *2026*
- **iF Design Award**, UX/UI, Samsung Vision AI *2026*
- **iF Design Award**, UX, WonderScope *2023*
- **National Merit Scholarship for Science and Engineering** | KSAF *2018*
- **Highest Academic Achievement Scholarship** | POSTECH *2018*

RESEARCH & PROFESSIONAL EXPERIENCE

Immersive Visualization &Interaction Lab, Georgia Tech *Oct. 2026 – Present (on-site)*
Visiting Researcher (Advisor: Prof. Yalong Yang) *May – Oct. 2026 (remote)*
Atlanta, Georgia

- Led *Spatial Acts as Prompts*, an XR interaction technique that uses spatial actions as implicit prompts, enabling AI to infer evolving intent during sensemaking [6].
- Built end-to-end Unity XR system and LLM inference pipeline, decomposing spatial behavior into arrangement (WHAT), movement dynamics (HOW), temporal context (WHEN) to infer sensemaking phases and deliver phase-aligned AI assistance.
- Validated phase inference through ablation studies (78.6% accuracy vs. 25.0% for static layouts); within-subject study (N=14) against voice prompting revealed how Spatial Act Prompting supports evolving plans through ongoing interaction.

Samsung Electronics, Seoul R&D Center *Jan. 2024 – Oct. 2026*
Research Engineer/Designer *Seoul, Korea*

- Investigated context-aware conversational AI interfaces for Gemini-powered *Vision AI Companion*, adapting response layouts and follow-up actions to user intent and on-screen content; contributed to global deployment; inventor on 10 AI interaction patent applications [P1-4].
- Analyzed KPIs across 1.4M+ devices to identify long-term interaction patterns and refine AI interfaces.
- Proposed and implemented *Vibe Hopping*, a web-based video interface extending semantic search beyond verbal queries through direct selection of scenes and objects, using Segment Anything (SAM).

WonderLab, KAIST *Mar. 2021 – Aug. 2023*
Research Assistant (Advisor: Prof. Woohun Lee) *Daejeon, Korea*

- Led research on desk-grounded XR interaction for immersive authoring [5], transforming 2D GUI into spatially multiplexed 3D controls with passive haptic feedback; evaluated impact on workflow pattern against raycasting (N=14).
- Developed body-driven VR interactions for room-scale spatial transformations [4], deriving gesture mappings from studies of spatial perception and implementing collision-based arm-swing slicing.
- Built two near-surface AR applications end-to-end, using sensor-fusion tracking to respond to hand movements over physical exhibits' surfaces; both demoed at ACM SIGGRAPH [3], with one deployed at the National Science Museum.

HCI Lab, School of Computing, KAIST *Jan. 2022 – Apr. 2022*
Visiting Student (Advisior: Prof. Geehyuk Lee, Prof. Woohun Lee) *Daejeon, Korea*

- Led interaction research for fisheye camera-based around-device input, defining pose-and-motion gesture vocabularies and deriving interaction principles for smart lamps and earbuds through formative studies.

MXXR, *UX Intern* *Nov. 2020 – Mar. 2021*
Seoul, Korea

- Designed onboarding interactions for mobile AR app with camera-based spatial sensing.

LG CNS, IT Research Institute, CTO Division *Jun. 2018 – Aug. 2018*
Software Engineering Intern *Seoul, Korea*

- Developed an Android smartwatch app for factory task monitoring, adapting its UI to workers' proximity and task states using BLE beacon sensing and Node.js/Socket.IO backend.

ONGOING RESEARCH

[Co-Lead] *Independent Collaboration with CMU Researchers* | ***CMU XRTC Creative Research Grant***
PickyRoom: A Room-Scale Agent for Steering Behavior Through Spatial Transformations

- Developing an LLM-driven XR agent framework that guides users toward their goals by selecting spatial transformations to modulate action costs, using shader manipulation and mesh generation.

[Lead] *Research at IVI Lab, Georgia Tech*
Action-Aware Situated Generative Widgets

- Investigating AR glasses interfaces that infer evolving user goals from physical actions and situational context to generate interactive widgets across everyday activities and applications.

ACADEMIC ACTIVITIES

EXHIBITION & PRESENTATION

- Presenter, CHI Interactivity Demo — *Japan 2025*
- Presenter, CHI Student Game Competition — *Germany 2023*
- Presenter, SIGGRAPH Emerging Technologies Demo — *Canada 2022*
- Selected Poster Presenter, SIGGRAPH Art Papers Roundtable — *Canada 2022*
- Research Featured on KBS, MBC, TJB (National Broadcasting) — *2022*
- Research Exhibitor, Korea National Science Museum Special Exhibition — *2022*
- Research Exhibitor, Korea National Science Museum Living Lab — *2021*
- Research Exhibitor, Gwacheon National Science Museum — *2021*

SERVICE & TEACHING

- Paper Reviewer (CHI 2026, CHI 2027), Student Volunteer (TEI 2022)
- Teaching Assistant, Design Entrepreneurship (KAIST ID402)

SELECTED PATENTS

[P4] **Yeeun Shin**, YoonYoung Kim, et al. “User Interface for Objectifying Generative AI Responses and Cross-Device Context Transfer” (*US · WO · KR P20260024357; Pending*)

[P3] **Yeeun Shin**, Celine Park, et al. “Interface and Operation Method for Context-Aware Real-Time Chat Archiving” (*US · PCT/KR2026/010885 · KR P20250108290; Pending*)

[P2] **Yeeun Shin**, Celine Park, et al. “Method for Domain Sorting and Generative Interface Provision Based on User Utterances” (*US 19/692041 · PCT/KR2026/004941 · KR P20250109125; Pending*)

[P1] YoonYoung Kim, **Yeeun Shin**, et al. “Real-Time Reactive Card Interface Based on Content and User Preferences” (*US US19/765904 · PCT/KR2026/007091 · KR P20250122239; Pending*)

SKILLS

Programming	Python, C#, C++, JavaScript/TypeScript, LLM APIs, FastAPI, Node.js, HTML, CSS, Git
Prototyping	Unity, Meta Quest SDK, XR Interaction Toolkit, Blender, Arduino, Raspberry Pi, Digital Fabrication, Rhino/Grasshopper, Figma, Sketch, Adobe CC
Research	**(quantitative)** Ablation Studies, Log Analysis, Statistical Analysis (Python, SPSS, SQL) **(qualitative)** Semi-structured Interviews, Thematic Analysis, Contextual Inquiry